

From linear algebra to knot theory via categories

Ross Street

Macquarie University

Mathematics Department Colloquium, E7B100

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- ▶ to define some concepts of monoidal category theory and identify them for vector spaces;
- ▶ to explain string notation as a method for calculating;
- ▶ to explain the two-way interaction between knot theory and categories.

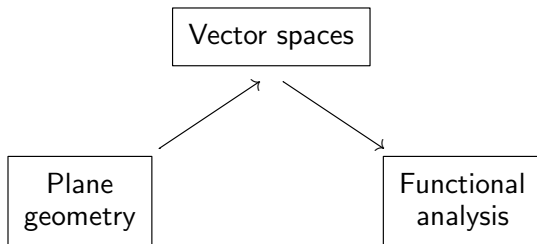
Role of monoidal categories in mathematics

Simplistically:

$$\frac{\text{Point in Euclidean Space}}{\text{Vector Space}} = \frac{\text{Vector Space}}{\text{Monoidal Category}} .$$

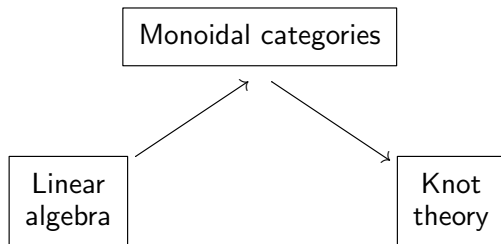
Left-hand side

Formalizing properties of the Euclidean plane into the structure of vector space, we are able to transfer low-dimensional thinking to obtain precise results in higher dimensions.



Right-hand side

Formalizing properties of the totality of vector spaces into the structure of monoidal category, we are able to transfer ideas from linear algebra into some perhaps surprising areas.



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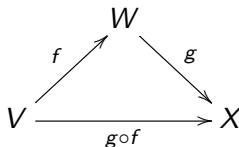
- ▶ Felix Klein (1872) recognized the importance of structure-respecting transformations in geometry. They are what define the specific geometry in question.
- ▶ These transformations were taken to be invertible, forming a groupoid.
- ▶ General structure-respecting functions are also crucial.
- ▶ The notion of “category” is a mathematical structure which captures these ideas.
- ▶ The original defining paper of Samuel Eilenberg and Saunders Mac Lane (1945) wanted to formalize what was meant by “naturally” or “canonically” isomorphic constructions in mathematics, particularly in group theory.

The category of vector spaces

- ▶ The *category* $\mathbf{Vect}$ of (say, complex) vector spaces consists not only of the vector spaces V (*objects*) but also of the linear functions $f : V \longrightarrow W$ between them (*morphisms*).

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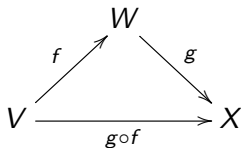
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of the morphisms.

- ▶ Each object V has an identity morphism $1_V : V \longrightarrow V$.

Operations on vector spaces

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- ▶ Bilinear functions $h : V \times W \longrightarrow X$ are in canonical bijection with linear functions $f : V \otimes W \longrightarrow X$.

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- ▶ with identities $1_A : A \longrightarrow A$,
- ▶ subject to associativity and identity axioms.

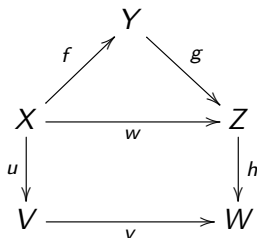
$$\begin{array}{ccccc}
 A & \xrightarrow{h \circ (g \circ f)} & D \\
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$$\begin{array}{ccccc}
 A & \xrightarrow{f} & B \\
 f = 1_B \circ f \downarrow & \nearrow 1_B & \downarrow g = g \circ 1_B \\
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 \end{array}$$

Commutative diagrams

In a category, we can speak of *commutative diagrams*:



$$h \circ w = v \circ u \quad \text{and} \quad g \circ f = w, \quad \text{so } v \circ u = h \circ g \circ f.$$

Functors

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- ▶ A *functor* $T : \mathcal{C} \longrightarrow \mathcal{H}$ assigns an object TA of $\mathcal{H}$ to each object A of $\mathcal{C}$, a morphism $Tf : TA \longrightarrow TB$ in $\mathcal{H}$ to each $f : A \longrightarrow B$ in $\mathcal{C}$, such that

$$T1_A = 1_{TA} \quad \text{and} \quad T(g \circ f) = Tg \circ Tf.$$

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defined by

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- ▶ In an obvious sense of *product category*, tensor product is a functor

$$- \otimes - : \mathbf{Vect} \times \mathbf{Vect} \longrightarrow \mathbf{Vect}.$$

Other examples of functors

- ▶ There is a functor $T_n : \mathbf{Vect} \longrightarrow \mathbf{Vect}$ taking V to the n -fold tensor power

$$T_n V = \underbrace{V \otimes V \otimes \cdots \otimes V}_n = V^{\otimes n}.$$

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$$TV = \mathbb{C} \oplus V \oplus V^{\otimes 2} \oplus V^{\otimes 3} \dots$$

- ▶ In fact, each TV is an algebra: indeed, we have a functor

$$T : \mathbf{Vect} \longrightarrow \mathbf{Alg} .$$

Natural transformations

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- ▶ Categories exhibit a higher-dimensional structure in that there are morphisms between functors.
- ▶ Suppose S and $T : \mathcal{C} \longrightarrow \mathcal{H}$ are functors. A *natural transformation*

$$\theta : S \Longrightarrow T$$

is a family of morphisms

$$\theta_A : SA \longrightarrow TA$$

in $\mathcal{H}$, indexed by the objects A of $\mathcal{C}$, such that the square

$$\begin{array}{ccc} SA & \xrightarrow{\theta_A} & TA \\ Sf \downarrow & & \downarrow Tf \\ SB & \xrightarrow{\theta_B} & TB \end{array}$$

commutes for all $f : A \longrightarrow B$ in $\mathcal{C}$.

Examples of natural transformations

- ▶ Each linear function $t : V \longrightarrow W$ determines a natural transformation $t \otimes - : V \otimes - \Longrightarrow W \otimes -$ with components

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- ▶ The componentwise image of s gives the symmetric n -th-power functor $S_n : \mathbf{Vect} \longrightarrow \mathbf{Vect}$.

Monoidal category data

A *monoidal category* consists of a category $\mathcal{C}$ equipped with a functor

$$- \otimes - : \mathcal{C} \times \mathcal{C} \longrightarrow \mathcal{C},$$

an object I , and invertible natural families of morphisms

$$\alpha_{A,B,C} : (A \otimes B) \otimes C \longrightarrow A \otimes (B \otimes C)$$

$$\lambda_A : I \otimes A \longrightarrow A, \quad \rho_A : A \longrightarrow A \otimes I$$

such that the following diagrams commute.

Monoidal category axioms

$$\begin{array}{ccc}
 & (A \otimes (B \otimes C)) \otimes D & \\
 \alpha_{A,B,C} \otimes 1_D \nearrow & & \searrow \alpha_{A,B \otimes C,D} \\
 ((A \otimes B) \otimes C) \otimes D & & A \otimes ((B \otimes C) \otimes D) \\
 \alpha_{A \otimes B,C,D} \downarrow & & \downarrow 1_A \otimes \alpha_{B,C,D} \\
 (A \otimes B) \otimes (C \otimes D) & \xrightarrow{\alpha_{A,B,C \otimes D}} & A \otimes (B \otimes (C \otimes D))
 \end{array}$$

$$\begin{array}{ccc}
 (A \otimes I) \otimes B & \xrightarrow{\alpha_{A,I,B}} & A \otimes (I \otimes B) \\
 \rho_A \otimes 1_B \uparrow & & \downarrow 1_A \otimes \lambda_B \\
 A \otimes B & \xrightarrow{1_{A \otimes B}} & A \otimes B
 \end{array}$$

Example

Take $\mathcal{C} = \mathbf{Vect}$, usual tensor product of vector spaces as $\otimes$, and $I = \mathbb{C}$.

String diagrams

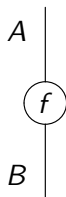
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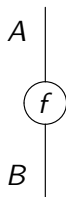
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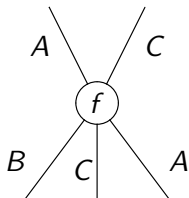
- ▶ Both notations are “1-dimensional”.

String diagrams (continued)

- ▶ In a monoidal category $\mathcal{C}$, we use two dimensions: composition is up-down while tensoring is left-right.

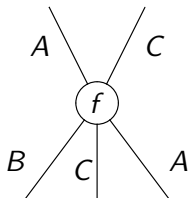
String diagrams (continued)

- ▶ In a monoidal category $\mathcal{C}$, we use two dimensions: composition is up-down while tensoring is left-right.
- ▶ Instead of $A \otimes C \xrightarrow{f} B \otimes C \otimes A$, we draw.

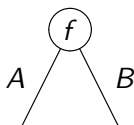


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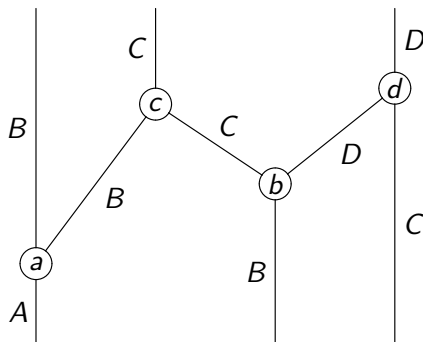


- ▶ A morphism $I \xrightarrow{f} A \otimes B$ is drawn.



Progressive plane string diagrams

- ▶ Notice that the monoidal structural items $\otimes, I, \alpha, \rho, \lambda$ do not appear.
- ▶



A diagram Γ as above, labelled in $\mathcal{C}$, is called a *progressive plane string diagram*. It means that we have morphisms

$$B \otimes B \xrightarrow{a} A, \quad C \otimes D \xrightarrow{b} B, \quad C \xrightarrow{c} B \otimes C, \quad D \xrightarrow{d} D \otimes C.$$

The value of a string diagram

- ▶ The *value* of the diagram is the composite

$$\begin{aligned} v(\Gamma) = & (B \otimes C \otimes D \xrightarrow{1_B \otimes c \otimes d} B \otimes B \otimes C \otimes D \otimes C \\ & \xrightarrow{1 \otimes 1 \otimes b \otimes 1} B \otimes B \otimes B \otimes C \xrightarrow{a \otimes 1 \otimes 1} A \otimes B \otimes C). \end{aligned}$$

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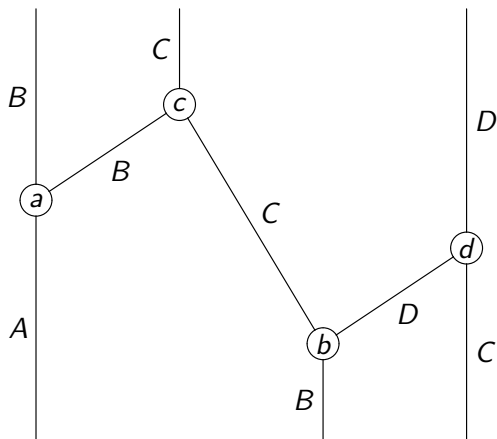
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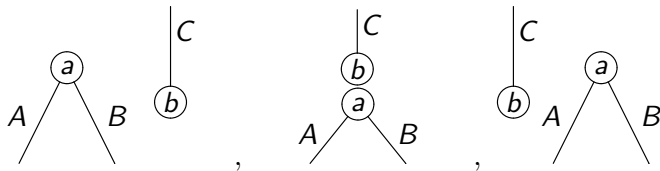
- ▶ It is obtained by composing the more elementary values of horizontal layers.
- ▶ This value is independent of *deformation* in a natural sense.
- ▶ For example, the following diagram is a deformation of Γ and has the same value as can be shown by the axioms for a monoidal category.



Its value is

$$v(\Gamma) = (B \otimes C \otimes D \xrightarrow{1_B \otimes c \otimes 1} B \otimes B \otimes C \otimes D \xrightarrow{a \otimes 1 \otimes 1} A \otimes C \otimes D \\ \xrightarrow{1 \otimes 1 \otimes d} A \otimes C \otimes D \otimes C \xrightarrow{1 \otimes b \otimes 1} A \otimes B \otimes C).$$

The notation handles units well: if $I \xrightarrow{a} A \otimes B$ and $C \xrightarrow{b} I$, then the following three string diagrams all have the same value.



Dual objects

We can look at some concepts which make sense in any monoidal category.

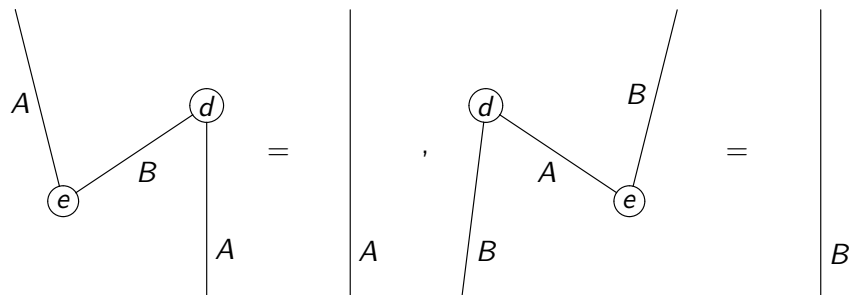
A *duality* between objects A and B is a pair of morphisms

$$e : A \otimes B \longrightarrow I \quad \text{and} \quad d : I \longrightarrow B \otimes A$$

called the *counit* and the *unit*, respectively, satisfying

$$\begin{array}{ccccc}
 A \otimes I & \xrightarrow{1 \otimes d} & A \otimes (B \otimes A) & \xrightarrow{\alpha^{-1}} & (A \otimes B) \otimes A & \xrightarrow{e \otimes 1} & I \otimes A \\
 & \searrow \rho_A^{-1} & & & & \nearrow \lambda_B & \\
 & & B & & & & \\
 \\
 I \otimes B & \xrightarrow{d \otimes 1} & (B \otimes A) \otimes B & \xrightarrow{\alpha} & B \otimes (A \otimes B) & \xrightarrow{1 \otimes e} & B \otimes I \\
 & \searrow \lambda_B & & & & \nearrow \rho_B^{-1} & \\
 & & B & & & &
 \end{array}$$

Duality axioms in string notation



Write $A \dashv B$ when such e and d exist:

A is a *left dual* for B while B is a *right dual* for A .

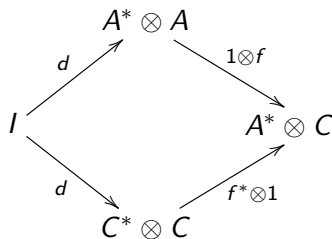
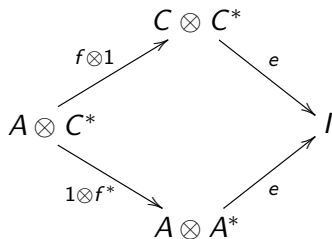
Any two right duals for A are isomorphic (if they exist).

Write A^* for a chosen right dual for A .

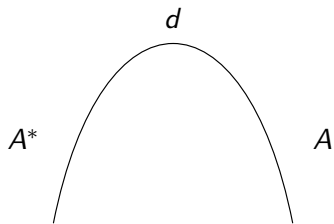
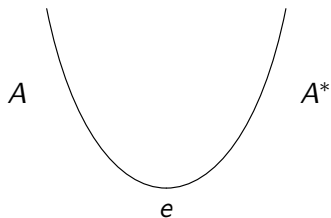
Call $\mathcal{C}$ right *autonomous* (“compact”, “rigid”) when each object has a right dual. Choice gives a functor

$$(\)^* : \mathcal{C}^{\text{op}} \longrightarrow \mathcal{C}$$

where “op” means $(\)^*$ is “contravariant”: $A \xrightarrow{f} C$ goes to $A^* \xleftarrow{f^*} C^*$.



We suppress e and d from notation by writing them as

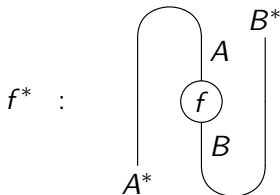


The duality conditions become string straightening rules.

$$\begin{array}{c} A \\ | \\ \text{loop} \\ | \\ A \end{array} = \begin{array}{c} | \\ A \end{array}, \quad \begin{array}{c} A^* \\ | \\ \text{loop} \\ | \\ A^* \end{array} = \begin{array}{c} | \\ A^* \end{array}$$

The diagram shows two string straightening rules. The first rule shows a vertical line labeled A at the top, which loops around a vertical line labeled A^* at the bottom, and then continues down to a vertical line labeled A at the bottom. This is equal to a single vertical line labeled A at the bottom. The second rule shows a vertical line labeled A^* at the top, which loops around a vertical line labeled A at the bottom, and then continues down to a vertical line labeled A^* at the bottom. This is equal to a single vertical line labeled A^* at the bottom.

This allows anticlockwise looping by taking $(\)^*$. f^* is depicted as follows.



Diagrams are not necessarily “progressive” any more.
They can backtrack.

Example

An object of $\mathbf{Vect}$ has a right (left) dual if and only if it is finite dimensional (exercise!).

If V has basis $v_1, \dots, v_n$ and $v_1^*, \dots, v_n^*$ is the “dual basis” of $V^* = \mathbb{C}^V$ then we have

$$V \otimes V^* \xrightarrow{e} \mathbb{C}$$

$$v \otimes \varphi \longmapsto \varphi(v)$$

$$\mathbb{C} \xrightarrow{d} V^* \otimes V$$

$$1 \longmapsto \sum_{i=1}^n v_i^* \otimes v_i.$$

Braidings

A *braiding* on a monoidal category $\mathcal{C}$ is a natural family of invertible morphisms

$$\gamma_{A,B} : A \otimes B \xrightarrow{\cong} B \otimes A$$

satisfying (ignoring α)

$$\begin{array}{ccc} A \otimes B \otimes C & \xrightarrow{\gamma_{A \otimes B, C}} & C \otimes A \otimes B \\ & \searrow 1 \otimes \gamma_{B, C} \quad \nearrow \gamma_{A, C} \otimes 1 & \\ & A \otimes C \otimes B & \end{array}$$

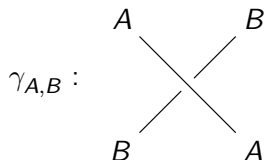
$$\begin{array}{ccc} A \otimes B \otimes C & \xrightarrow{\gamma_{A, B \otimes C}} & B \otimes C \otimes A \\ & \searrow \gamma_{A, B} \otimes 1 \quad \nearrow 1 \otimes \gamma_{A, C} & \\ & B \otimes A \otimes C & \end{array}$$

Example

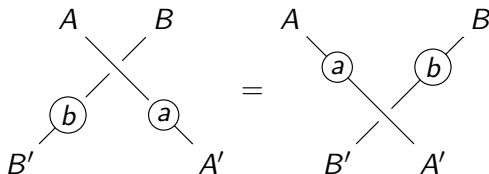
In $\mathbf{Vect}$, $\gamma_{A,B}(a \otimes b) = b \otimes a$.

Enter three dimensions!

By writing γ as a *crossing*



our string diagrams move into three-dimensional Euclidean space.
The axioms say



$$\begin{array}{c} A \quad B \quad C \\ \diagdown \quad \diagup \quad \diagdown \\ \diagup \quad \diagdown \quad \diagup \end{array} = \begin{array}{c} \text{ } \\ \diagdown \quad \diagup \quad \diagdown \\ \diagup \quad \diagdown \quad \diagup \\ \text{ } \end{array} \quad , \quad \begin{array}{c} A \quad B \quad C \\ \diagdown \quad \diagup \quad \diagdown \\ \diagup \quad \diagdown \quad \diagup \end{array} = \begin{array}{c} \text{ } \\ \diagdown \quad \diagup \quad \diagdown \\ \diagup \quad \diagdown \quad \diagup \\ \text{ } \end{array} .$$

$C \quad A \quad B$

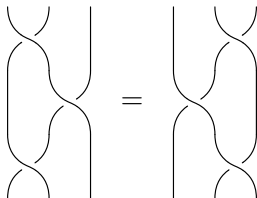
 $B \quad C \quad A$

A monoidal category with chosen braiding is called *braided*.

The braiding automatically satisfies what is called the *Yang–Baxter equation* or the *braid identity* or a *Reidemeister move* in different contexts. Here is the proof:

$$\begin{array}{ccccc}
 & & B \otimes A \otimes C & \xrightarrow{1 \otimes \gamma} & B \otimes C \otimes A \\
 & \nearrow \gamma \otimes 1 & & \nearrow \gamma_{A, B \otimes C} & \\
 A \otimes B \otimes C & & & & \\
 & \searrow 1 \otimes \gamma & & \searrow \gamma_{A, C \otimes B} & \\
 & A \otimes C \otimes B & \xrightarrow{\gamma \otimes 1} & C \otimes A \otimes B & \\
 & & & \nearrow 1 \otimes \gamma & \\
 & & & C \otimes B \otimes A &
 \end{array}$$

naturality

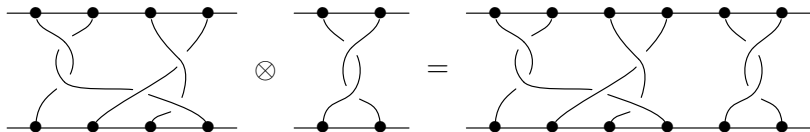


The braid category $\mathbb{B}$

Here is an example of a braided monoidal category which is vastly different from $\mathbf{Vect}$. Objects are natural numbers $0, 1, 2, \dots$

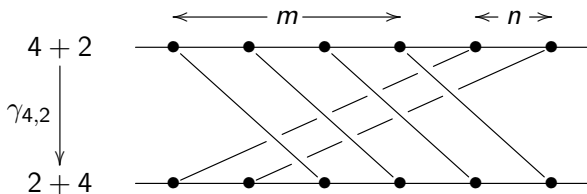
$$\mathbb{B}(m, n) = \begin{cases} \emptyset & \text{when } m \neq n \\ \text{braid group } B_n \text{ on } n \text{ strings} & \text{when } m = n \end{cases}$$

The functor $\mathbb{B} \times \mathbb{B} \xrightarrow{\otimes} \mathbb{B}$ is defined on objects by $(m, n) \mapsto m + n$ and on morphisms as indicated by:



The isomorphisms α, λ, ρ are identities.

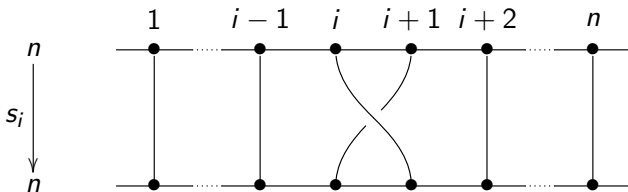
The braiding $\gamma_{m,n}$ is indicated by:



Proposition

$\mathbb{B}$ is the free braided monoidal category generated by a single object.

The single generating object is 1 and tensoring the braiding $\gamma_{1,1}$ on both sides with identity morphisms gives the generators s_i for the braid group B_n as indicated:



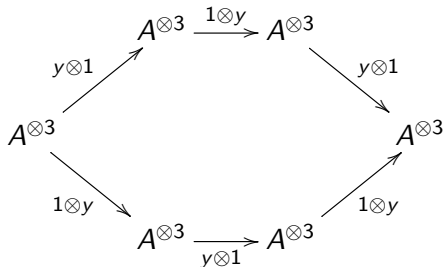
The Proposition means that there is an equivalence of categories

$$\text{BStMon}(\mathbb{B}, \mathcal{C}) \simeq \mathcal{C}$$

between the category of braiding-and-tensor-preserving functors $\mathbb{B} \rightarrow \mathcal{C}$ and the category $\mathcal{C}$ itself, for every braided monoidal category $\mathcal{C}$.

Yang-Baxter operators

A *Yang-Baxter operator* $y : A \otimes A \rightarrow A \otimes A$ on an object A of any monoidal category $\mathcal{C}$ is an invertible morphism satisfying commutativity of the following hexagon. There is an obvious category $\mathbf{YB}\mathcal{C}$ of such pairs (A, y) .



If $\mathcal{C}$ is braided, each object A has a Yang-Baxter operator on it, namely,

$$y = \gamma_{A,A} : A \otimes A \longrightarrow A \otimes A.$$

Proposition

$\mathbb{B}$ is the free monoidal category generated by an object bearing a Yang–Baxter operator. That is, for every monoidal category $\mathcal{C}$,

$$\mathrm{StMon}(\mathbb{B}, \mathcal{C}) \simeq \mathrm{YB}\mathcal{C}.$$

Proposition

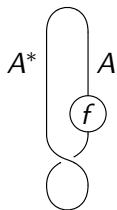
Any right autonomous braided monoidal category is left autonomous.

To see this we take $A^* \otimes A \xrightarrow{\gamma} A \otimes A^* \xrightarrow{e} I$ and $I \xrightarrow{d} A^* \otimes A \xrightarrow{\gamma^{-1}} A \otimes A^*$ as counit and unit for A^* as left dual to A .

Trace of an endomorphism

The *trace* of an endomorphism $f : A \longrightarrow A$ in an autonomous braided monoidal category is defined by

$$\mathrm{Tr}(f) = \left(I \xrightarrow{d} A^* \otimes A \xrightarrow{1 \otimes f} A^* \otimes A \xrightarrow{e'} I \right).$$



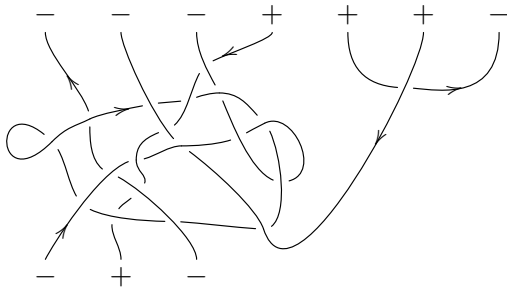
Example

If $V \in \mathbf{Vect}$ is finite dimensional with basis $v_1, \dots, v_n$ and f is defined by $f(v_i) = \sum_j a_{ij} v_j$, then

$$\begin{aligned}\mathrm{Tr}(f) : \mathbb{C} &\longrightarrow \mathbb{C} \\ 1 &\longmapsto \sum_i a_{ii} .\end{aligned}$$

The category $\mathbb{T}$ of tangles

This is a monoidal category first defined by Yetter (1988).
The objects are words $- + + - - +$ in symbols $+$ and $-$.
Morphisms are tangles



Freyd-Yetter (1989) showed that $\mathbb{T}$ is an autonomous braided monoidal category with a “freeness property”.

We can see that $\mathbb{B}$ is the braided monoidal subcategory of $\mathbb{T}$ by identifying $n \in \mathbb{B}$ with $\underbrace{+ \dots +}_n \in \mathbb{T}$.

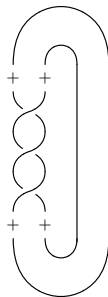
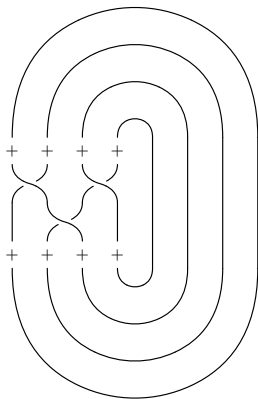
The braids occur as the progressive endomorphisms in $\mathbb{T}$.

The endomorphisms of the unit object 0 are precisely the *oriented links*.

Proposition

The trace of a braid in $\mathbb{T}$ is the link obtained as the Markov closure.

Here are examples of the trace of a braid, the second one is a trefoil.



Invariants of links

Each tensor-preserving (= strong monoidal) functor

$$F : \mathbb{T} \longrightarrow \mathbf{Vect}$$

will produce a complex number for each oriented link ℓ :

$$0 \xrightarrow{\ell} 0 \longmapsto \mathbb{C} \xrightarrow{F\ell} \mathbb{C}.$$

Such F are determined by a single vector space V and a special kind of Yang–Baxter operator

$$y : V \otimes V \cong V \otimes V,$$

on V :

$$F(+ + - + -) = V \otimes V \otimes V^* \otimes V \otimes V^*.$$

There are techniques for solving the Yang–Baxter equation to find such functors, and hence, invariants of links; see Turaev (1988).

Trace without duals

A (*right*) *internal hom* C^A for objects A and C of a monoidal category $\mathcal{C}$ is an object equipped with a morphism (called “evaluation”)

$$\mathrm{ev} : A \otimes C^A \longrightarrow C$$

which induces a bijection

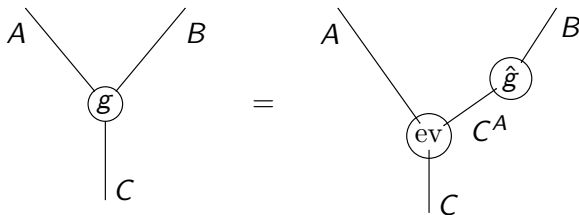
$$\begin{aligned} \mathcal{C}(B, C^A) &\cong \mathcal{C}(A \otimes B, C) \\ (B \xrightarrow{f} C^A) &\longmapsto (A \otimes B \xrightarrow{1 \otimes f} A \otimes C^A \xrightarrow{\mathrm{ev}} C). \end{aligned}$$

If A has a left dual $A^\vee$ then $C^A \cong A^\vee \otimes C$.

Yet internal homs can exist without duals (e.g., in $\mathcal{C} = \mathbf{Vect}$ and $\mathbf{Set}$).

Trace in analysis

I know of no extension of the string notation, in keeping with the geometry as for duals, which covers internal homs.



From field theory in physics there are geometric situations where internal homs and trace can exist without duals.

This led Stolz-Teichner (arXiv:1010.4527) to develop a notion of *trace* using only the “weak duals” I^A .

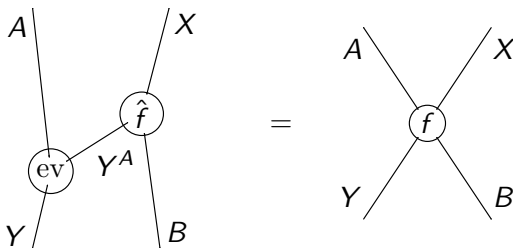
The following treatment of “contraction” is adapted from these authors.

Approximability and nuclearity

A quadruple (X, Y, A, B) of objects of a monoidal category $\mathcal{C}$ is called *approximable* when Y^A exists and the following composite is injective.

$$\mathcal{C}(X, Y^A \otimes B) \xrightarrow{A \otimes -} \mathcal{C}(A \otimes X, A \otimes Y^A \otimes B) \xrightarrow{\mathcal{C}(1, \text{ev} \otimes 1)} \mathcal{C}(A \otimes X, Y \otimes B)$$

A morphism $f : A \otimes X \longrightarrow Y \otimes B$ is called *nuclear* when it is in the image of that composite.



More general trace

Example

If A has a right dual then any (X, Y, A, B) is approximable. The appropriate composite is invertible so every f is nuclear.

The *trace* (or *contraction by A*) of

$$f : A \otimes X \longrightarrow Y \otimes A$$

in a braided monoidal $\mathcal{C}$ is the composite

$$\mathrm{Tr}(f) : X \xrightarrow{\hat{f}} Y^A \otimes A \xrightarrow{\gamma} A \otimes Y^A \xrightarrow{\mathrm{ev}} Y.$$

Let $\mathbf{TV}$ be the category of topological complex vector spaces which are locally convex, complete and Hausdorff.

The tensor product represents continuous bilinear maps.

Theorem (Stolz-Teichner)

Take $\mathcal{C} = \mathbf{TV}$.

1. *If A has the approximation property (from analysis) then $(\mathbb{C}, \mathbb{C}, A, A)$ is approximable.*
2. *A morphism $f : A \longrightarrow B$ is nuclear as in analysis if and only if $f : A \otimes \mathbb{C} \longrightarrow \mathbb{C} \otimes B$ is nuclear.*
3. *If A has the approximation property and $f : A \longrightarrow A$ is nuclear then the trace $\mathrm{Tr}(f)$ is the classical trace.*